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12 - STD	HALF YEARLY EXAMINATION - 2025	
TIME :3.00 Hrs	CHEMISTRY	MARKS : 70

PART - I 17.12.25

15 x 1 = 15

Choose the correct answer:

1. The most effective electrolyte for the coagulation of As_2S_3 sol is
(a) NaCl (b) $Al_2(SO_4)_3$ (c) $Ba(NO_3)_2$ (d) $K_3[Fe(CN)_6]$

2. Which of the metal is extracted by Hall-Heroult process?
(a) Al (b) Ni (c) Cu (d) Zn

3. The compound prepared by Clemmensen reduction reaction is
a) Alcohol (b) Aldehyde (c) Alkane (d) Alkene

4. Glucose $\xrightarrow{HCN}$ Product $\xrightarrow{Hydrolysis}$ Product $\xrightarrow{HI/\Delta}$ A. The compound A is:
(a) Heptanoic acid (b) 2-Iodohexane (c) Heptane (d) Heptanol

5. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.1 M HCl are mixed?
(a) 2.0 (b) 3 (c) 7.0 (d) 12.65

6. $HO\text{-}CH_2\text{-}CH_2\text{-}OH$ on heating with periodic acid gives
(a) Methanoic acid (b) Glyoxal (c) Methanol (d) CO_2

7. **Assertion:** 2-methyl-1,3-butadiene is the monomer of natural rubber.
Reason: Natural rubber is formed through anionic addition polymerisation.
(a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) Assertion is true but reason is false. (d) Both assertion and reason are false.

8. The electrode which has less tendency to undergo corrosion
(a) Zn^{2+}/Zn (b) Cu^{2+}/Cu (c) Al^{3+}/Al (d) Pb^{4+}/Pb^{2+}

9. The hybridization of carbon atoms in fullerene (C_{60}) is
(a) sp^3 (b) sp (c) sp^2 (d) Partially sp^2 and Partially sp^3

10. Which of the following lanthanoid ions is diamagnetic?
(a) Eu^{2+} (b) Yb^{2+} (c) Ce^{2+} (d) Sm^{2+}

11. Match the following

(1) AgBr	(i) Metal excess defect
(2) ZnO (heat)	(ii) F Center
(3) FeO	(iii) Metal deficiency defect
(4) NaCl (in presence of Na vapour)	(iv) Frenkel defect

(a) (1)-(ii), (2)-(iv), (3)-(i), (4)-(iii) (b) (1)-(iv), (2)-(i), (3)-(iii), (4)-(ii)
(c) (1)-(iv), (2)-(ii), (3)-(iii), (4)-(i) (d) (1)-(ii), (2)-(iv), (3)-(iii), (4)-(i)

12. The product formed by the reaction of an aldehyde with a primary amine
(a) Carboxylic acid (b) Aromatic acid (c) Schiff's base (d) Ketone

13. Which of the following reactions has a half-life independent of the initial concentration of reactants?
(a) Zero order reaction (b) First order reaction
(c) Pseudo first order reaction (d) None of these

14. The basicity of pyrophosphorous acid ($H_4P_2O_5$) is
(a) 4 (b) 2 (c) 3 (d) 5

15. Which type of isomers is exhibited by $[Pt(NH_3)_2Cl_2]$ - Geometrical Isomerism.

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15. Which type of isomerism is exhibited by $[Pt(NH_3)_2Cl_2]$?
(a) Coordination isomerism (b) Linkage isomerism
(c) Optical isomerism (d) Geometrical isomerism

PART - II

Answer Any Six Questions. (Q. No 24 Is Compulsory) **6 x 2 = 12**

16. Give the uses of Borax.
17. Why do zirconium and hafnium exhibit similar properties?
18. Define unit cell.
19. Explain common ion effect with an example.
20. State Kohlrausch Law.
21. Give the basic requirement for vapour phase refining.
22. What is electrophoresis?
23. Write a note on vulcanization of rubber.
24. Find A and B.

$$CH_3NO_2 \xrightarrow{LiAlH_4} A \xrightarrow{CH_3Br} B$$

PART - III

Answer Any Six Questions. (Q. No 33 Is Compulsory) **6 x 3 = 18**

25. Explain Schottky defect.
26. What is catenation? Describe briefly the catenation property of carbon.
27. Give the differences between double salts and co-ordination compounds.
28. What type of hybridization occurs in (a) BrF_5 (b) XeF_4 (c) XeO_3?
29. State Faraday's Laws of electrolysis.
30. How do you convert phenol into salicylic acid?
31. Formic acid acts as a reducing agent - why? Prove it with any one test.
32. Give any three differences between DNA and RNA.
33. The rate constant for a first order reaction is $1.54 \times 10^{-3}\ s^{-1}$. Calculate its half lifetime.

PART - IV

Answer All The Questions. **5 x 5 = 25**

34. (a) Explain Zone refining process with an example. **(OR)**
(b) (i) How will you identify borate radical?
(ii) What are the effects of Lanthanoide contraction?
35. (a) What are the main assumptions of valence bond theory? **(OR)**
(b) (i) Derive Integrated rate law for a zero order reaction A → Product.
(ii) Explain pseudo first-order reaction with an example.
36. (a) What are the characteristics of catalysts? **(OR)**
(b) (i) Differentiate crystalline solids and amorphous solids.
(ii) Define buffer Index.
37. (a) How does anisole react with the following reagents?
(i) CH_3COCl/anhydrous $AlCl_3$
(ii) Conc H_2SO_4/Conc HNO_3
(iii) Br_2/CH_3COOH **(OR)**
(b) (i) What is Urotropine? How is it prepared?
(ii) Why aniline does not undergo Friedel Crafts reaction?
38. (a) Explain the structure of Glucose. **(OR)**
(b) (i) How will you distinguish 1°, 2° and 3° alcohol by Lucas method?
(ii) What are antacids?

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